IAG DENUCHANGE WORKSHOP · GREECE 2026

An integrated graph theory and remote sensing approach to functional sediment connectivity analysis in an Alpine proglacial area across multiple temporal scales

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1 Types of sediment connectivity

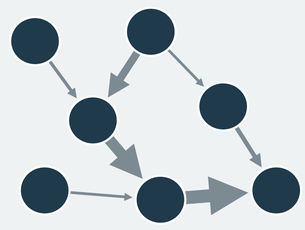
Rockwalls, moraines, hillslopes, glacio-fluvial deposits, terraces and the fluvial corridor are linked by sediment pathways. These links can be seen in two ways:

Structural connectivity
The spatial configuration of landforms: how they are physically arranged and linked

Functional connectivity
How sediment actually moves at a given point in time: highly dynamic and difficult to quantify

2 Graph theory for functional connectivity

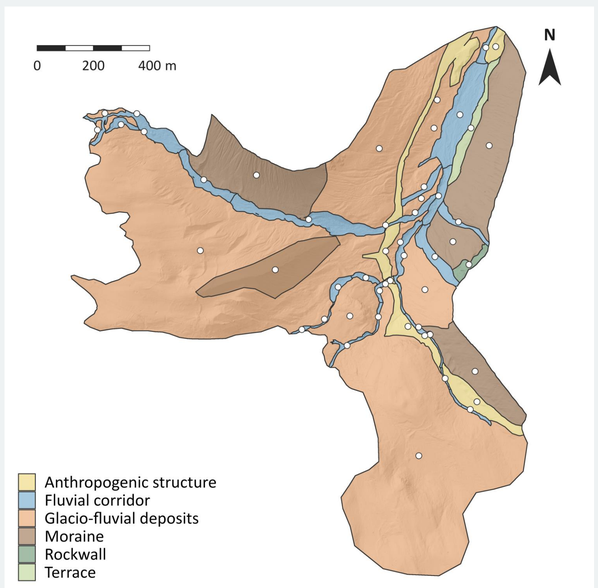
- Graph theory is the maths of networks: nodes joined by edges
- A framework to quantify functional connectivity
- Landforms become nodes; sediment pathways become edges
- Edges show whether landforms are connected; edge weights indicate how much sediment moves



node = landform
edge width = sediment flux

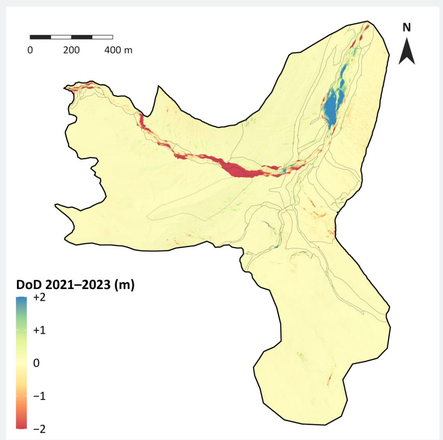
3 Mapping the Sulden proglacial region, Italian Alps

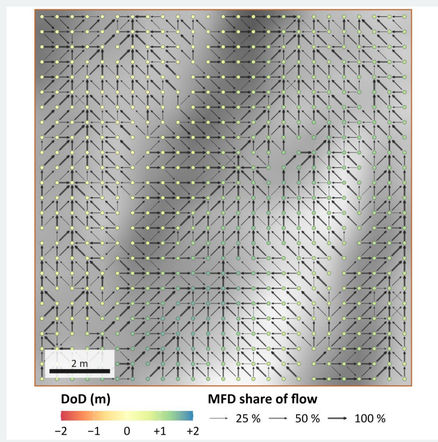
- **This study:** functional connectivity between 2021 and 2023
- **Data:** 2021 LiDAR DEM and 2023 UAV photogrammetric DEM, both at 0.5 m resolution
- Landforms mapped on the 2021 DEM, the start of the study period
- **Every landform is a graph node**



4 Modelling sediment pathways

- **DoD:** local changes in each pixel (erosion or deposition), 2021-23
- **MFD:** where each pixel passes material on, and in what proportions
- **Accumulated DoD:** DoD summed over each pixel's contributing area

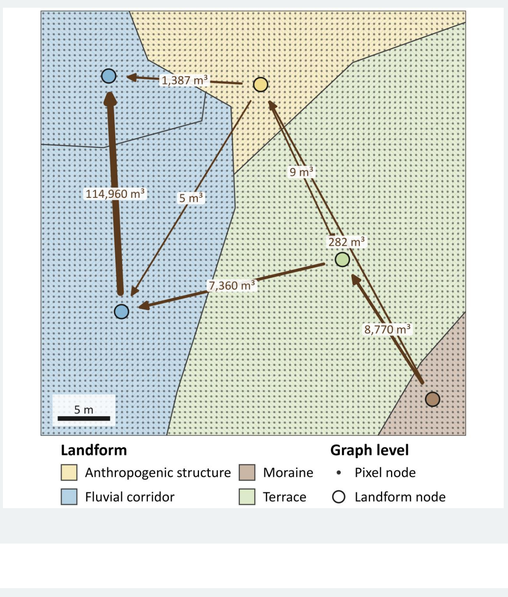




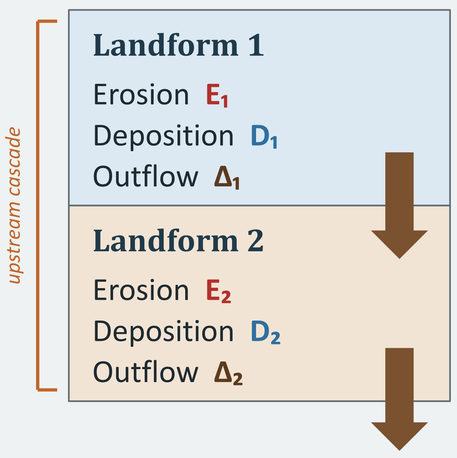
5 From pixel level to landform level

- Pixels within each mapped unit merge into one landform node
- Pixel links between two landforms are summed into one weighted edge

pixel level → landform level
Nodes **12,685,368 → 51**
Links **11,847,703 → 116**



6 Quantifying functional connectivity



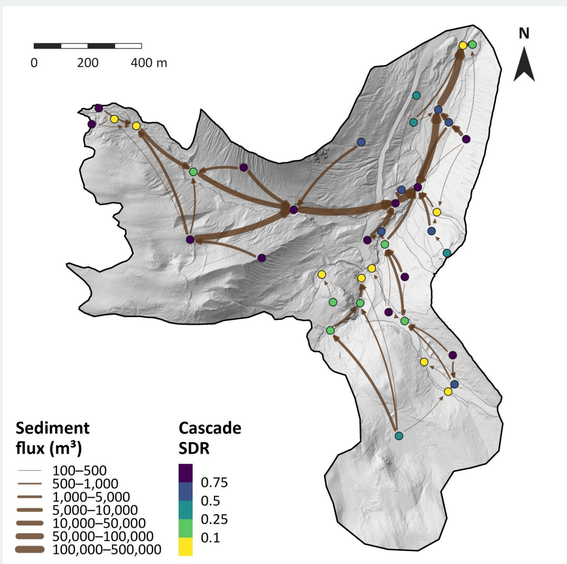
The sediment delivery ratio (SDR) is a metric that quantifies functional connectivity.

$\Delta_1 = E_1 - D_1$
$\Delta_2 = E_1 + E_2 - D_2$

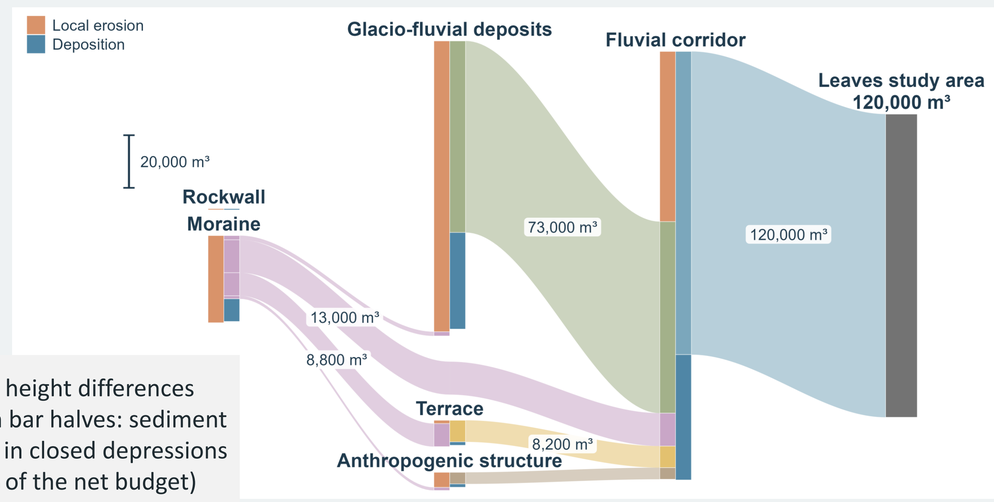
Cascade SDR
= outflow ÷ all erosion upstream
$\mathbf{SDR_{cascade} = \Delta_2 \div (E_1 + E_2)}$

7 The final functional connectivity graph

- The western Sulden proglacial area is the most active in transferring sediment, as shown by the thicker arrows
- **Quality check:** Upslope-directed arrows are absent or insignificant; they could be minimized by using MFD during landform delineation (Section 3)



8 Fluxes between landforms



Small height differences between bar halves: sediment retained in closed depressions (3.8% of the net budget)

9 Key takeaway

→ Viewing landforms as graphs offers a new perspective on sediment connectivity and the potential to uncover previously hidden patterns.

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